

Daksh — Class X Study Test

Subject: Mathematics | **Test:** Test 4 (Day 28, 6 June 2026) — 2-Week Cumulative
Time Allowed: 3 Hours • Maximum Marks: 80

Syllabus covered: Chs 1–13 (comprehensive): Real Numbers • Polynomials • Linear Equations • Quadratic Equations • AP • Triangles • Coordinate Geometry • Trigonometry • Applications • Circles • Areas Related to Circles • Surface Areas and Volumes • Statistics

General Instructions:

- (i) This question paper contains five sections — A, B, C, D and E.
- (ii) Section A has Multiple Choice / Very Short Answer Questions of 1 mark each.
- (iii) Section B has Short Answer Questions of 2 marks each.
- (iv) Section C has Short Answer Questions of 3 marks each.
- (v) Section D has Long Answer Questions of 5 marks each.
- (vi) Section E has Case-based / Source-based questions of 4 marks each.
- (vii) All questions are compulsory. Internal choice has been provided in some questions wherever indicated.
- (viii) Use of calculators is not permitted.

SECTION A — Multiple Choice / Very Short Answer Questions (20 × 1 = 20 marks)

- Q1.** If two positive integers $a = pq^2$ and $b = p^3q$, where p, q are primes, then $\text{LCM}(a, b) = ?$ (a) pq (b) p^2q^2 (c) p^3q^2 (d) p^2q
- Q2.** The decimal expansion of $13/3125$ is: (a) terminating (b) recurring (c) non-terminating non-recurring (d) cannot be determined
- Q3.** If α and β are zeroes of $x^2 - p(x + 1) - c$ such that $(\alpha + 1)(\beta + 1) = 0$, then $c = ?$ (a) 0 (b) 1 (c) -1 (d) 2
- Q4.** The pair $x - 3y = 7$ and $3x - 9y = 21$ represents: (a) parallel lines (b) intersecting lines (c) coincident lines (d) perpendicular lines
- Q5.** The roots of $x^2 - 2x - 8 = 0$ are: (a) 4, -2 (b) $-4, 2$ (c) 4, 2 (d) $-4, -2$
- Q6.** The 11th term of the AP: $-3, -1/2, 2, \dots$ is: (a) 22 (b) 22.5 (c) 25 (d) 25.5
- Q7.** The point of intersection of the lines $x = 4$ and $y = -3$ is: (a) $(4, -3)$ (b) $(-3, 4)$ (c) $(-4, 3)$ (d) $(3, -4)$
- Q8.** The distance of the point $(-6, 8)$ from origin is: (a) 14 (b) 100 (c) 10 (d) 8
- Q9.** $\tan 60^\circ \cdot \cot 30^\circ = ?$ (a) 1 (b) $\sqrt{3}$ (c) 3 (d) $1/3$
- Q10.** If $\sin \theta = 3/5$, then $\cos \theta = ?$ (a) $3/4$ (b) $4/5$ (c) $5/4$ (d) $3/5$
- Q11.** The angle of elevation of the top of a tower 100 m high from a point at a distance of 100 m on level ground is: (a) 30° (b) 45° (c) 60° (d) 90°
- Q12.** The length of the tangent drawn from a point 13 cm away from the centre of a circle of radius 5 cm is: (a) 8 cm (b) 12 cm (c) 18 cm (d) $\sqrt{169}$ cm
- Q13.** The area of a sector of a circle of radius 14 cm and central angle 90° is: (a) 154 cm^2 (b) 308 cm^2 (c) 77 cm^2 (d) 49 cm^2
- Q14.** The volume of a hemisphere of radius 7 cm is: (a) $(2/3)\pi(7)^3$ (b) $(4/3)\pi(7)^3$ (c) $\pi(7)^3$ (d) $(1/2)\pi(7)^2$
- Q15.** If a cone, a hemisphere and a cylinder stand on equal bases and have the same height, the ratio of their volumes is: (a) $1 : 2 : 3$ (b) $2 : 1 : 3$ (c) $3 : 2 : 1$ (d) $1 : 3 : 2$
- Q16.** For data with mean 18 and median 20, the mode is: (a) 22 (b) 24 (c) 16 (d) 20
- Q17.** Find the LCM of 16 and 24.
- Q18.** Find the value of k for which the system $kx - y = 2$ and $6x - 2y = 3$ has no solution.
- Q19.** If $\sec A = 17/8$, find $\tan A$.
- Q20.** If the perimeters of two similar triangles are in the ratio $4 : 5$, find the ratio of their corresponding sides.

SECTION B — Short Answer-I (5 × 2 = 10 marks)

- Q21.** Find the largest number which divides 615 and 963 leaving the remainder 6 in each case.
- Q22.** Find the value of k such that $x^2 + kx + 4 = 0$ has equal roots.

Q23. Find the 31st term of an AP whose 11th term is 38 and 16th term is 73.

Q24. Two tangents PA and PB are drawn from an external point P to a circle with centre O. Prove that $\angle APB + \angle AOB = 180^\circ$.

Q25. If the points A(6, 1), B(8, 2), C(9, 4) and D(p, 3) are vertices of a parallelogram, taken in order, find the value of p.

SECTION C — Short Answer-II ($6 \times 3 = 18$ marks)

Q26. Prove that $2 + 3\sqrt{5}$ is irrational, given that $\sqrt{5}$ is irrational.

Q27. If two zeroes of polynomial $x^3 - 6x^2 - 26x - 35$ are $2 \pm \sqrt{3}$, find the other zeroes.

Q28. A motorboat whose speed is 18 km/h in still water takes 1 hour more to go 24 km upstream than to return downstream to the same spot. Find the speed of the stream.

Q29. Show that any positive odd integer is of the form $6q + 1$, $6q + 3$ or $6q + 5$, where q is some integer.

Q30. Prove the identity: $(\cos A - \sin A + 1)/(\cos A + \sin A - 1) = \operatorname{cosec} A + \cot A$.

Q31. Find the area of the sector of a circle of radius 5 cm whose central angle is 60° . Hence find the area of the corresponding minor segment. (Use $\pi = 3.14$, $\sqrt{3} \approx 1.73$)

SECTION D — Long Answer ($4 \times 5 = 20$ marks)

Q32. The angles of elevation of the top of a tower from two points at distances a and b metres from the base of the tower and in the same straight line with it are complementary. Prove that the height of the tower is $\sqrt{ab}$ metres.

Q33. Prove that 'the lengths of tangents drawn from an external point to a circle are equal'. Use this to deduce that the parallelogram circumscribing a circle is a rhombus.

Q34. A solid is in the form of a cylinder with hemispherical ends. The total length of the solid is 35 cm, and the diameter of the cylinder is 14 cm. Find the total volume and total surface area of the solid.

Q35. Find the mean and median of the following frequency distribution. Marks: 0–10, 10–20, 20–30, 30–40, 40–50; Number of students: 8, 12, 25, 9, 6.

SECTION E — Case-based / Source-based ($3 \times 4 = 12$ marks)

Q36. Case Study (Statistics — Survey). A survey was conducted in a city about the number of children per family. The data is summarised below:

Children per family 0–11–22–33–44–5
Number of families 51125104
(i) What is the modal class? (1)
(ii) Find the mode of the data. (2)
(iii) What is the most common number of children per family? (1)

Q37. Case Study (Trigonometry — Heights & Distances). An aeroplane is flying at a height of 5 km above the ground. At a particular instant, the angle of elevation of the aeroplane from a point on the ground is 60° . After 10 seconds, the angle of elevation becomes 30° .

(i) Draw a labelled diagram of the situation. (1)
(ii) Find the horizontal distance covered by the aeroplane in 10 seconds. (3) (Use $\sqrt{3} \approx 1.73$)

Q38. Case Study (Surface Areas and Volumes — Capsule). A medicine capsule is in the shape of a cylinder with two hemispheres stuck to each of its ends. The length of the entire capsule is 14 mm and its diameter is 5 mm.

(i) What is the radius of the hemispheres? (1)
(ii) What is the height of the cylindrical part of the capsule? (1)
(iii) Find the surface area of the capsule. (Use $\pi = 22/7$) (2)